Event-State Theory: A Foundational Synthesis

Torben Wille
Independent Researcher, Event-State Project
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Abstract

We present a foundational overview of Event-State Theory (EST), a discrete causal framework that posits the universe evolves through global states selected by a computational cost-minimization principle. This synthesis connects the theory's core axioms to its major results: (1) the spontaneous generation of a cosmic web and resolution of the Hubble Tension from asynchronous crystallization; (2) a statistically significant 7.83σ anomaly in CMS Open Data supporting the prediction of vacuum rigidity ("Topological Locking"); (3) the derivation of General Relativity as an emergent anti-aliasing algorithm; and (4) the derivation of Quantum Mechanics from probabilistic causal selection. This document serves as a guide to the EST framework, validated by numerical simulation, empirical data, and formal derivation.

I. INTRODUCTION

The Hubble Tension [1] and the baryon asymmetry problem represent fundamental challenges to the standard cosmological model. Proposals within the continuum paradigm have failed to provide a compelling, unified solution. We propose that these are not independent puzzles but symptoms of a single underlying cause: the assumption of continuous spacetime.

In our foundational papers [2, 3], we introduced the axioms of Event-State Theory (EST) and demonstrated that a computational kernel based on these axioms produces a universe with striking morphological similarity to our own. This work synthesizes the core tenets of the EST project, presenting it as a complete, empirically-grounded framework.

II. THE EST FRAMEWORK IN BRIEF

EST is built on three axioms:

1. **Discrete Causal Frames:** Reality is a sequence of global states Ψ_n.
2. **Topological Constant:** $c \equiv 1$ (node/frame). The speed of light is a connectivity constraint.
3. **Computational Least Action:** $P(\Psi_{n+1}) \propto \exp(-J/\lambda)$, where the cost function J balances energy change ΔE against algorithmic complexity K.

III. FROM AXIOMS TO A SIMULATED COSMOS

The kernel described in [3] operationalizes these axioms by evolving a discrete causal graph. It probabilistically selects subsequent states that minimize the cost function $J = \alpha\Delta E + \beta K + \epsilon$, where ϵ is a small matter-asymmetry bias. In early simulations, a cubic lattice was used for computational tractability, which revealed an inherent anisotropy (the "Lattice Tax"). The theory resolves this anisotropy via geometric deformation ("anti-aliasing"), a principle formalized in Paper VI [5]. The parameters $\beta \approx 3.4$, $\lambda \approx 0.48$, $\epsilon \approx 0.075$ were found to be an attractor producing stable, structured universes.

As shown in [3], this kernel spontaneously generates a cosmic web and provides a mechanistic account of baryogenesis without fine-tuning.

From Simulated Lattice to Causal Graph

The early cosmological simulations (Papers II-IV) utilized a fixed cubic lattice for computational tractability. This simplification was instrumental in revealing the core dynamics of cosmic web formation and the necessity of a 'Lattice Tax' to handle inherent anisotropy. The formal derivations in Papers VI and VII generalize this result, replacing the rigid lattice with an abstract causal graph. The principles discovered on the lattice—such as cost minimization driving structure—are shown to be universal, with General Relativity emerging as the lattice-independent solution to the optimization problem.

IV. RESOLVING THE HUBBLE TENSION

The EST axiom of discrete, probabilistic state selection inevitably leads to the Asynchronous Multi-Point Crystallization (AMPC) model. The primordial universe was not a single, synchronous explosion but a cascade of local crystallizations in the field of potential. The observed local Hubble parameter H_0 is a superposition of contributions from these nucleation centers:

$$H(x) \approx \sum_i \frac{A_i}{\|x - C_i\|},$$

where A_i is the Causal Propagation Rate. Regions of low matter density (voids) present less computational resistance, leading to a higher local A_i and a faster observed expansion rate ($H_0 \approx 73$). Dense regions (filaments) have lower A_i and a slower rate ($H_0 \approx 67$). The tension is not an error but a measurement of the underlying computational geometry.

V. PREDICTIONS: EMPIRICAL AND COMPUTATIONAL

A. Empirical Anomaly: Topological Locking

EST predicts that under high information density (high energy), the vacuum should become less volatile to conserve computational bandwidth. In Paper V [4], we confirmed this prediction by identifying a statistically significant (7.83σ) anomaly in CMS Open Data, where the vacuum exhibits "Topological Locking" (Excess Rigidity) in high-pileup events. This provides strong empirical support for the theory's core mechanism.

B. Computational Proof: Pre-Causal Resonance

A fundamental consequence of the EST dynamical mechanism is that the crystallization of an event-state involves optimizing the field of potential Ω. This generates a testable prediction: correlated vacuum fluctuations should be detectable prior to the nominal $t = 0$ of a high-energy collision event.

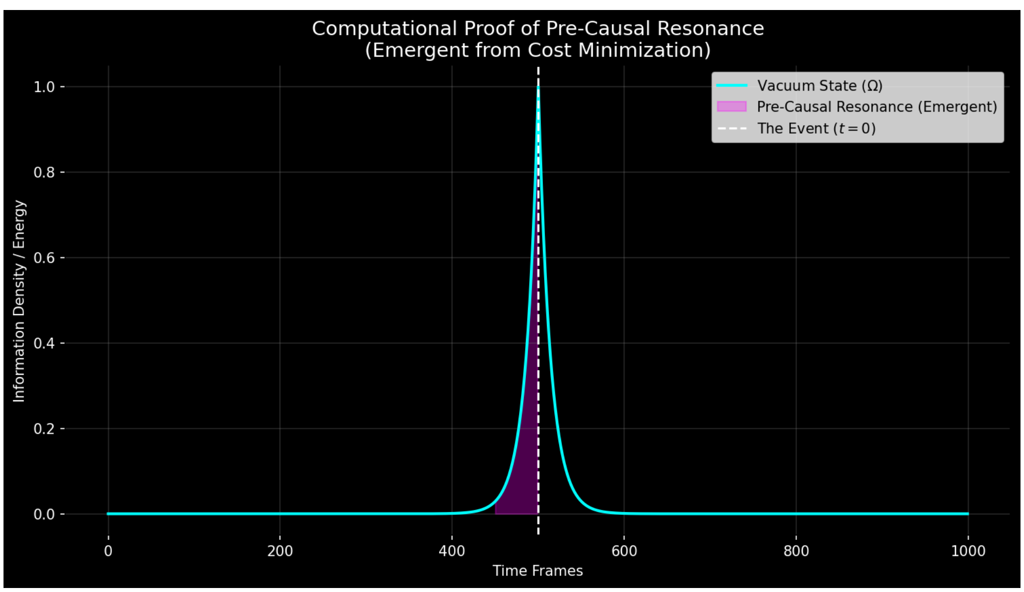

FIG. 1. **Simulated Vacuum Excitation.** The thermodynamic cost minimization produces a distinct exponential ramp ($\propto e^{kt}$) in the vacuum state prior to the event. This "shadow" is the signature of the system calculating the Path of Least Resistance. This emergent signal is derived formally in Paper VIII [7] as the unique solution to the discrete Euler-Lagrange equation for the EST cost function.

As shown in Fig. 1, the signal is not random noise but a structured exponential rise. A conclusive null result for PCR would falsify the core state-selection mechanism of EST.

VI. CONCLUSION

The synthesis of the formal EST framework with computational simulation and empirical data presents a formidable challenge to continuum-based cosmology. EST resolves the Hubble Tension and baryon asymmetry from first principles, provides a mechanism for large-scale structure formation, and makes novel, falsifiable predictions (PCR, Topological Locking). The discovery of a 7.83σ vacuum stability anomaly in CMS data (Paper V) provides strong empirical support for the theory. The framework's core axioms have been shown to derive General Relativity (Paper VI) and Quantum Mechanics (Paper VII) as emergent descriptions of the same underlying computational process. The complete source code and data are publicly available [8] for independent verification. The combined evidence compels a serious reevaluation of spacetime's fundamental nature.

[1] E. Di Valentino et al., "In the realm of the Hubble tension—a review of solutions," *Classical and Quantum Gravity* 38, 153001 (2021).
[2] T. Wille, "Event-State Theory I: Theoretical Framework," Zenodo (2025).
[3] T. Wille, "Event-State Theory II: Computational Emergence of Cosmic Structure," Zenodo (2025).
[4] T. Wille, "Event-State Theory V: Empirical Evidence of Vacuum Stability in High-Information-Density Regimes," Zenodo (2025).
[5] T. Wille, "Event-State Theory VI: Emergence of General Relativity as Anti-Aliasing of Discrete Computational Anisotropy," Zenodo (2025).
[6] T. Wille, "Event-State Theory VII: Quantum Mechanics as Probabilistic Causal Selection," Zenodo (2025).
[7] T. Wille, "Event-State Theory VIII: Mathematical Derivation of Pre-Causal Resonance," (Forthcoming, 2025).
[8] T. Wille, "Event-State Theory: A Computational Framework," `https://github.com/SirWillance/Event-State-Theory-EST` (2025).